Confidence Interval Estimation

Range of values

Why do we need confidence interval?

Suppose the data is huge

↓

Draw random samples out of it

$S_1 \rightarrow \bar{x}_1$

$S_2 \rightarrow \bar{x}_2$

⋮ ⋮

} different means (we cannot trust on these values)

That's why we create a range to observe the population parameter → CI

Formula →

$$CI = \text{Point Estimate} \pm \text{Margin of Error}$$

where, point estimate = single value

MOE = how much this single value can be wrong

Ex- Find the avg. marks of students?

Given :- Point Estimate = 70

MOE = ±5

∴ CI = 70 ± 5

= 70 − 5 , 70 + 5

= (65 , 75)

* Steps to calculate Confidence Interval :-

① Point Estimate ⇝ sample mean ($\bar{x}$)

② Standard Error ⇝ $SE = \frac{s}{\sqrt{n}}$ (s = sample std)

③ Critical Value ⇝ A boundary point that separates the correct values from incorrect values

↓

Calculate using z-value / t-value

↙ (It is used when n>30)

↘ (It is used when n<30)

④ Margin of Error = Critical Value × SE

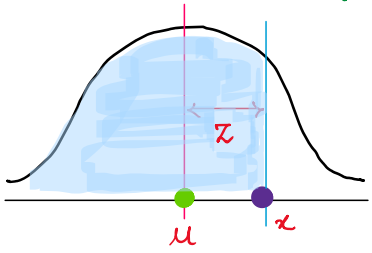
⑤ Confidence Interval = Point Estimate ± MOE

$$CI = \bar{x} \pm MOE$$

* Z- Score :-

It tells us how many standard deviations away a value is from the mean.

$$z = \frac{x - \mu}{\sigma}$$



Z- table :-

It is a mathematical table, indicating the values of the cumulative distribution function of the normal dist.

Standard Normal Probabilities

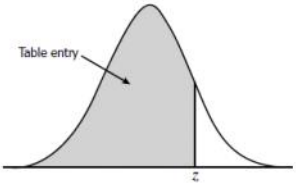


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177

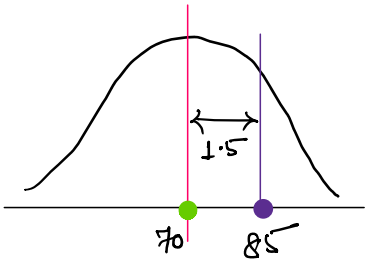
Ques. The marks obtained by 100 students in an exam are normally distributed with a mean of 70 & std of 10.
A student 'B' scored 85 marks.
Find → ① Z-score
② Percentage of students who scored less than 'B'

Given :- $\mu = 70$, $\sigma = 10$, $X = 85$, $n = 100$

so, $Z = \frac{X - \mu}{\sigma}$

$Z = \frac{85 - 70}{10}$

$Z = 1.5$ [This is z-score]



Now, Looking at the table with $Z = 1.5$

for $Z = 1.5$, the value obtained is :

$P(Z = 1.5) = 0.9332$ or 93.32%

∴ Ans → There are 93.32% students who scored marks below student 'B'

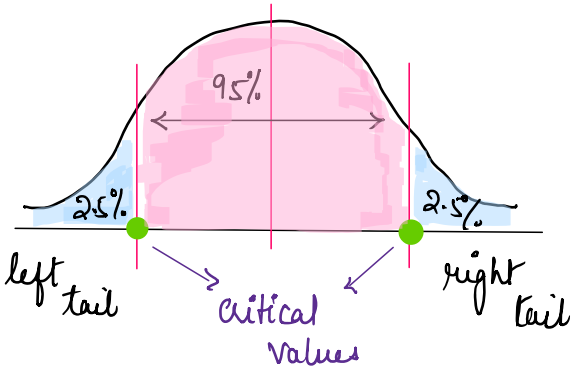
③ What is the percentage of students who scored above 85?

$P(Z > 1.5) = 1 - 0.9332$

$= 0.0668$ or 6.68%

④ What are the critical values for 95% CI?

95% of the middle data :-

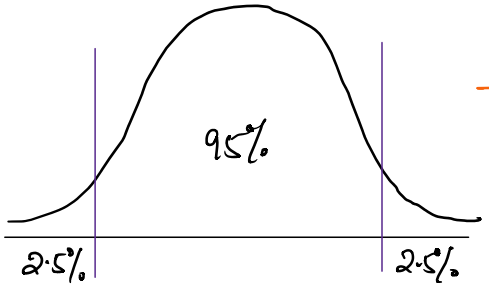


100 − 95 = 5%

As the data is symmetric

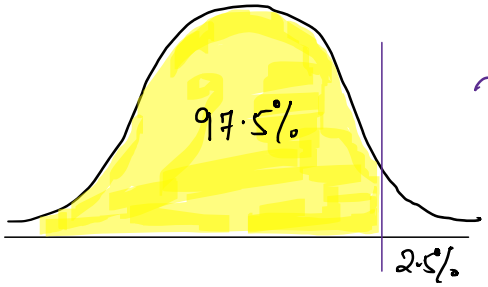
$\frac{5}{2} = 2.5\%$ or 0.025

How to calculate these critical boundaries?



→ Two-tail problem

For better understanding we make it a one-tail problem



⇝ How much area lie below the right tail?

Ans → 97.5%

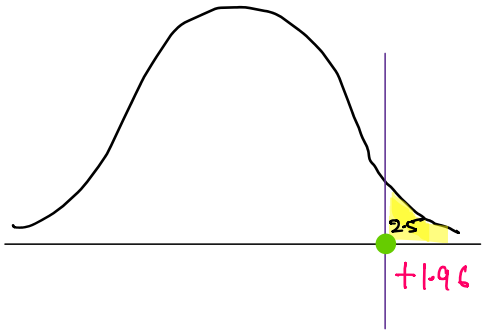
or

$1 - 0.025 = 0.975$

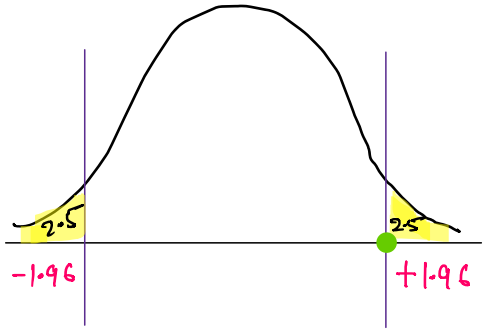
Here, we calculate the Z-score whose cumulative prob. is 0.975 using Z-table

∴ The exact match is → +1.96

So,



Similarly, calculate left critical value



∴ critical values → (-1.96, +1.96)

⑤ Calculate Confidence Interval

$$CI = PE \pm MOE$$

Here, $PE = \bar{x} = 70$

& $MOE = C.V. * SE$

∵ $SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$

So, $MOE = 1.96 * 1$

$MOE = 1.96$

& $CI = 70 \pm 1.96$

$CI = (70 - 1.96, 70 + 1.96)$

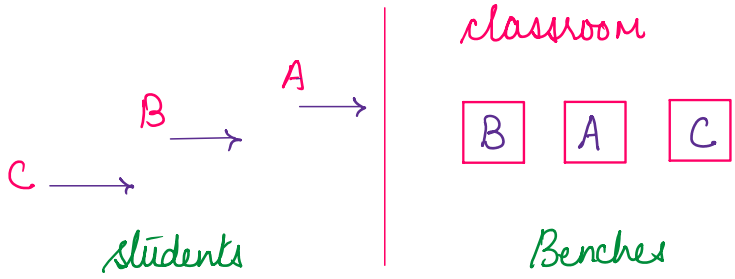
$CI = (68.04, 71.96)$

★ T-Statistics :-

↳ forms a distribution when $n<30$ [sample size is small]

↳ It also uses dof

DOF [Degree of Freedom] :-



Here, DOF (A) = 3

DOF (B) = 2

DOF (C) = 1

In t-statistics :-

Let $n = 10$

So, $dof = n-1$

$dof = 10-1 = 9$ ⇝ using this value we look at the t-Table to calculate critical value

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Ex- $n = 25$, Confidence level → 95% or 0.95

Not confident → 5% = 0.05 [1-tail]

2.5% = 0.025 [2-tail]

① dof = n-1

= 25-1

= 24

② looking at t-table with dof & not confident value

for 1-tail → 1.711

for 2-tail → 2.492

} critical values

★ Diff. b/w Normal Distribution & t-distribution :-

Normal Dist.	t- Dist.
• Bell shaped curve & symmetric with thinner tails	Bell shaped curve & symmetric with heavier tails
• Used when $n \geq 30$ & σ_{pop} is known	Used when $n < 30$ & σ_{pop} is unknown

